

GRAVESEND-BROADNESS, United Kingdom

WMO: 037840

Lat: 51.4645N

Lon: 0.3113E

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 96-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.2	-5.3	2.4	-0.2	-3.9	2.7	0.7	13.3	10.3	12.1	10.0	2.7	160	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.9	19.6	25.8	18.6	24.0	17.7	20.6	26.2	19.5	24.2	18.5	22.6	3.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.4	22.7	17.6	12.6	21.6	16.8	12.0	20.7	59.1	26.3	55.4	24.1	52.3	22.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	9.9	8.7	-5.4	31.9	1.7	2.1	-6.6	33.4	-7.6	34.6	-8.5	35.7	-9.7	37.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-5.7	22.5	1.9	1.7	-7.1	23.7	-8.2	24.7	-9.2	25.6	-10.5	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	5.5	5.7	7.5	9.9	13.0	16.2	18.4	18.3	15.6	12.3	8.5	5.9
	DBStd	5.53	3.34	3.18	3.00	3.00	2.81	2.57	2.45	2.41	2.40	2.75	3.09	3.49
	HDD10.0	604	142	123	88	38	6	0	0	0	0	11	63	133
	HDD18.3	2593	398	353	337	254	164	73	29	29	86	187	295	386
	CDD10.0	1131	3	3	9	34	101	187	261	258	169	82	18	6
	CDD18.3	78	0	0	0	0	1	9	33	29	6	1	0	0
	CDH23.3	550	0	0	0	3	13	86	218	189	36	4	0	0
	CDH26.7	117	0	0	0	0	1	19	41	50	5	1	0	0
Wind	WSAvg	4.3	4.9	5.0	4.7	4.3	4.4	4.0	3.8	3.8	3.9	4.2	4.3	4.6
Precipitation	PrecAvg	653	67	52	39	47	49	45	45	53	50	69	75	64
	PrecMax	933	160	118	102	125	120	117	97	96	126	182	169	132
	PrecMin	465	11	7	4	0	10	9	9	3	10	10	27	10
	PrecStd	116	35	31	23	32	25	33	20	26	29	39	37	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	14.2	17.6	22.9	24.9	29.2	30.0	30.6	27.1	21.8	16.7	14.4
		MCWB	11.7	11.2	12.4	15.7	18.2	19.8	20.3	20.7	18.7	17.4	14.9	12.6
	2%	DB	12.6	12.5	15.1	19.1	22.5	25.7	27.7	27.7	23.8	18.8	15.2	13.3
		MCWB	11.0	10.2	10.9	13.3	16.6	18.4	19.5	19.7	17.9	15.2	13.3	11.8
	5%	DB	11.5	11.5	13.6	16.9	20.5	23.4	25.8	25.2	21.7	17.6	14.2	12.4
		MCWB	9.9	9.4	10.2	11.9	15.3	17.1	18.5	18.5	16.8	14.7	12.3	10.9
	10%	DB	10.5	10.4	12.3	15.2	18.6	21.6	23.9	23.1	20.2	16.5	13.2	11.3
		MCWB	9.0	8.5	9.6	11.1	14.0	16.2	17.7	17.5	15.9	14.1	11.5	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.8	13.1	15.9	19.2	21.0	22.5	21.8	20.2	17.5	15.2	13.1
		MCDB	13.2	13.6	16.3	21.9	23.6	27.2	27.3	29.0	24.5	21.2	16.2	13.9
	2%	WB	11.3	10.7	11.7	13.9	17.3	19.2	20.6	20.4	18.7	16.2	13.7	12.1
		MCDB	12.4	12.1	14.2	17.8	21.3	23.8	25.9	26.1	22.4	18.0	14.9	13.2
	5%	WB	10.1	9.7	10.8	12.6	15.8	17.9	19.2	19.3	17.6	15.4	12.7	11.0
		MCDB	11.3	11.1	13.0	15.9	19.6	22.2	24.2	23.7	20.8	17.0	13.9	12.2
	10%	WB	9.2	8.8	9.8	11.6	14.6	16.7	18.2	18.3	16.7	14.6	11.6	9.8
		MCDB	10.4	10.2	11.9	14.5	17.9	20.8	22.7	22.0	19.3	16.1	13.0	11.1
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.6	7.0	8.7	9.0	9.2	9.4	9.0	8.6	7.0	5.6	4.8
		MCDBR	5.4	6.4	9.4	11.9	12.3	12.6	12.8	12.5	10.9	8.0	5.6	4.9
		MCWBR	5.1	4.8	6.0	6.7	7.2	6.6	6.0	5.6	5.9	5.2	4.6	4.5
	5% WB	MCDBR	5.3	5.8	8.5	10.8	11.2	11.4	11.5	11.1	9.9	7.0	5.4	4.9
		MCWBR	5.1	4.9	5.9	6.5	7.1	6.5	6.2	5.5	5.8	5.1	4.8	4.6
Clear-Sky Solar Irradiance	taub	0.334	0.349	0.387	0.407	0.413	0.414	0.411	0.401	0.382	0.365	0.338	0.324	
	taud	2.369	2.371	2.260	2.228	2.254	2.277	2.296	2.342	2.390	2.418	2.414	2.360	
	Ebn at Noon	670	768	800	829	841	842	838	827	801	739	662	616	
	Edh at Noon	63	82	109	126	129	127	123	112	95	77	60	55	
All-Sky Solar Radiation	RadAvg	0.79	1.46	2.58	4.11	4.90	5.44	5.16	4.30	3.23	1.85	0.95	0.61	
	RadStd	0.07	0.21	0.33	0.40	0.34	0.45	0.46	0.33	0.26	0.17	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page